# CSP-203: Software Tools & Technologies Lab

## Chapter – 4

## Plotting Tools: GNU Plot (Lecture – 1)

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**What is GNU Plot?**

- Gnuplot is a command-line and GUI-based plotting program that generates 2D and 3D plots of functions, data, and data fits

**About GNU Plot?**

- Released in 1986, it's widely used in scientific and academic fields for data visualization.
- Compatible with Linux, Windows, macOS, and other operating systems.
- Supports various formats like PNG, JPEG, SVG, PDF, and EPS, suitable for publications and presentations
- Highly scriptable, allowing automation and reproducibility of plots.

**Reason for using GNU plot for plotting:**

- Seamless Integration with Linux Tools: Gnuplot integrates well with Linux utilities and scripting languages, allowing for efficient data processing and visualization workflows.
- Scriptability: Its command-line interface enables automation through shell scripts, facilitating batch processing of multiple datasets.
- Wide Range of Output Formats: Gnuplot supports various output formats, including PNG, PDF, SVG, making it versatile for different presentation needs.
- Efficient Handling of Large Datasets: It is optimized for performance, capable of processing and plotting large volumes of data efficiently.
- Active Development and Support: Gnuplot has been under continuous development since 1986, ensuring regular updates and a supportive user community.

- **Installing GNU plot in linux:**
  1. sudo apt-get update
  2. sudo apt-get upgrade
  3. sudo apt-get install gnuplot
- **Checking the version of GNU plot:**

  gnuplot --version
- **How to run GNU plot program:**

  Just type: gnuplot on your terminal.
- **How to exit from GNU plot program:**

  Type: exit on your terminal

```
arindam@ARINDAM:~$ gnuplot --version
gnuplot 5.4 patchlevel 2
```

```
arindam@ARINDAM:~$ gnuplot

        G N U P L O T
        Version 5.4 patchlevel 2    last m

        Copyright (C) 1986-1993, 1998, 200
        Thomas Williams, Colin Kelley and

        gnuplot home:     http://www.gnupl
        faq, bugs, etc:   type "help FAQ"
        immediate help:   type "help"  (pl

Terminal type is now 'qt'
gnuplot> exit
arindam@ARINDAM:~$
```

❑ **Basic Arithmetic and Power Functions available in GNU plot:**

| Function | Description | Example |
|---|---|---|
| abs(x) | Absolute value | abs(-3) → 3 |
| sqrt(x) | Square root | sqrt(9) → 3 |
| exp(x) | Exponential | exp(1) → 2.718 |
| log(x) | Natural logarithm | log(2.718) |
| log10(x) | Base-10 logarithm | log10(1000) |
| int(x) | Truncate to integer (toward zero) | int(4.9) → 4 |
| ceil(x) | Ceiling (next highest integer) | ceil(4.2) → 5 |
| floor(x) | Floor (next lowest integer) | floor(4.9) → 4 |
| round(x) | Round to nearest integer | round(4.6) → 5 |
| sgn(x) / sign(x) | Sign function | sign(-3) → -1 |

## ❑ Trigonometric Functions available in GNU plot:

| Function | Description | Example |
|---|---|---|
| sin(x) | Sine | sin(pi/2) → 1 |
| cos(x) | Cosine | cos(0) → 1 |
| tan(x) | Tangent | tan(pi/4) → 1 |
| asin(x) | Arcsine (in radians) | asin(1) → π/2 |
| acos(x) | Arccosine | acos(1) → 0 |
| atan(x) | Arctangent | atan(1) → π/4 |
| atan2(y, x) | Arctangent of y/x (2-arg form) | atan2(1,1) → π/4 |

## ❑ Hyperbolic Functions available in GNU plot:

| Function | Description |
|---|---|
| sinh(x) | Hyperbolic sine |
| cosh(x) | Hyperbolic cosine |
| tanh(x) | Hyperbolic tangent |
| asinh(x) | Inverse hyperbolic sine |
| acosh(x) | Inverse hyperbolic cosine |
| atanh(x) | Inverse hyperbolic tangent |

## ❑ Random and Conditional Functions available in GNU plot:

| Function | Description | Example |
|---|---|---|
| rand(seed) | Random number in [0,1) | rand(0) |
| random(n) | Random integer in [0,n−1] | random(5) |
| defined(x) | Returns 1 if variable is defined | defined(foo) |
| exists("var") | Returns 1 if variable exists | exists("x") |

## ❑ Special Functions available in GNU plot:

| Function | Description | Example |
|---|---|---|
| gamma(x) | Gamma function Γ(x) | gamma(5) → 24 |
| lgamma(x) | Natural log of Gamma function | lgamma(5) |
| erf(x) | Error function | erf(1) |
| erfc(x) | Complementary error function | erfc(1) |
| norm(x) | Cumulative normal distribution | norm(0) → 0.5 |

## ❑ Complex Functions available in GNU plot:

| Function | Description | Example |
|---|---|---|
| real(z) | Real part | real(2+3i) → 2 |
| imag(z) | Imaginary part | imag(2+3i) → 3 |
| abs(z) | Magnitude | abs(3+4i) → 5 |
| arg(z) | Argument (angle) | arg(1+i) → π/4 |
| conj(z) | Complex conjugate | conj(3+4i) → 3-4i |
| re(z) | Alias for real(z) | |
| im(z) | Alias for imag(z) | |

## ❑ Constant Functions available in GNU plot:

| Constant | Value/Description |
|---|---|
| pi | π = 3.141592653589... |
| NaN | Not-a-Number |
| Inf | Infinity |

## ❑ Arithmetic Operators available in GNU plot:

| Operator | Description | Example | Result |
|---|---|---|---|
| + | Addition | 3 + 2 | 5 |
| - | Subtraction | 5 - 2 | 3 |
| * | Multiplication | 4 * 3 | 12 |
| / | Division | 10 / 2 | 5 |
| ** | Power (exponentiation) | 2 ** 3 | 8 |
| % | Modulus (remainder) | 7 % 3 | 1 |

- **Relational Operators available in GNU plot:**

| Operator | Description | Example | Result |
|---|---|---|---|
| == | Equal to | 5 == 5 | 1 (true) |
| != | Not equal to | 5 != 4 | 1 (true) |
| > | Greater than | 4 > 3 | 1 |
| < | Less than | 3 < 5 | 1 |
| >= | Greater than or equal | 5 >= 5 | 1 |
| <= | Less than or equal | 4 <= 4 | 1 |

- **Logical Operators available in GNU plot:**

| Operator | Description | Example | Result |
|---|---|---|---|
| && | Logical AND | (1 && 0) | 0 |
| ` | | ` | Logical OR |
| ! | Logical NOT | !1 | 0 |

## ❑ Assignement Operators available in GNU plot:

| Operator | Description | Example |
|---|---|---|
| = | Assign value to variable | a = 10 |
| += | Add and assign | a += 5 |
| -= | Subtract and assign | a -= 2 |
| *= | Multiply and assign | a *= 3 |
| /= | Divide and assign | a /= 2 |

## ❑ Ternary Operator available in GNU plot:

| Syntax | Description |
|---|---|
| condition ? value1 : value2 | If-then-else logic |

- **Variables:**
  - Storing in a variable:
    - a = 5 (Storing integer)
    - a = 4.789 (Storing float)
    - a = {3,4} ( 3+4i )
    - a = “STTL”
- **Printing variables:**
  - print “The store value of a is: ”, a

```
gnuplot> x = 2
gnuplot> print (x*x)
4
gnuplot>
```

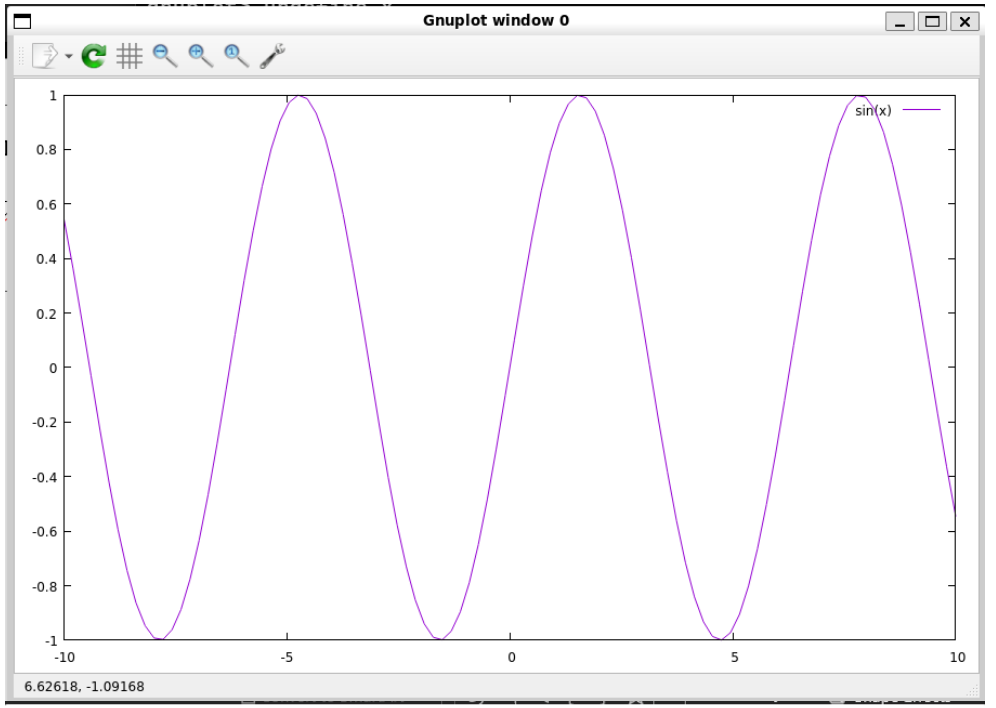
```
gnuplot> a = {3,5}
gnuplot> print "Real part of a is: ", real(a)
Real part of a is: 3.0
gnuplot>
```

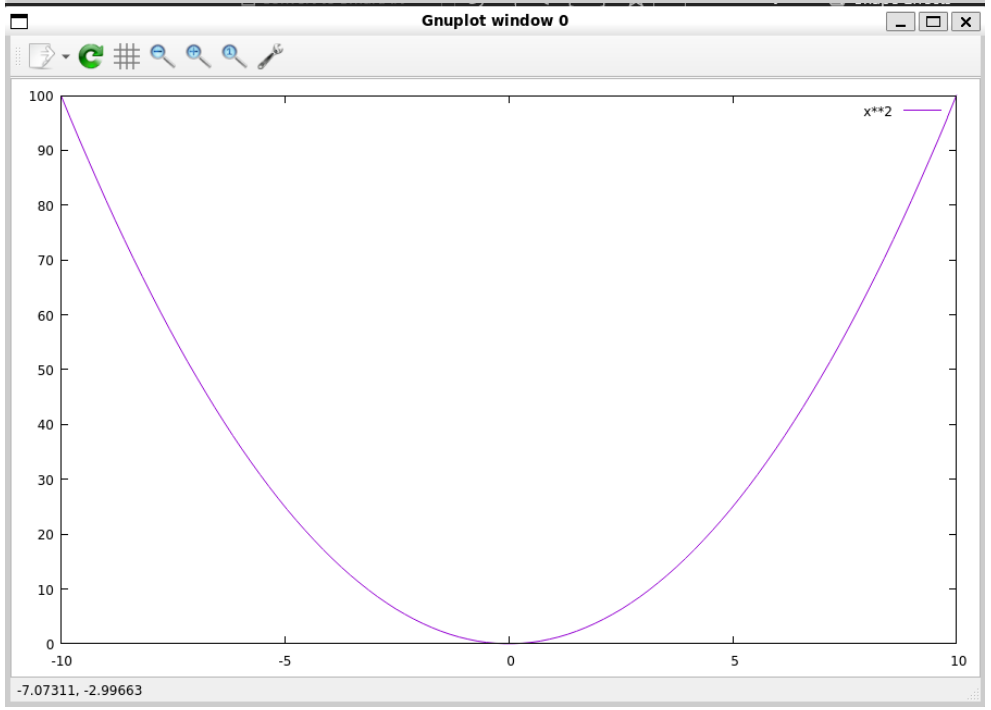
- **Clear Terminal in GNU plot:**
  - Command: !clear
- **Undefine variable:**
  - Command: undefine variable_name

```
gnuplot> a = 25
gnuplot> print a
25
gnuplot> undefine a
gnuplot> print a
         undefined variable: a

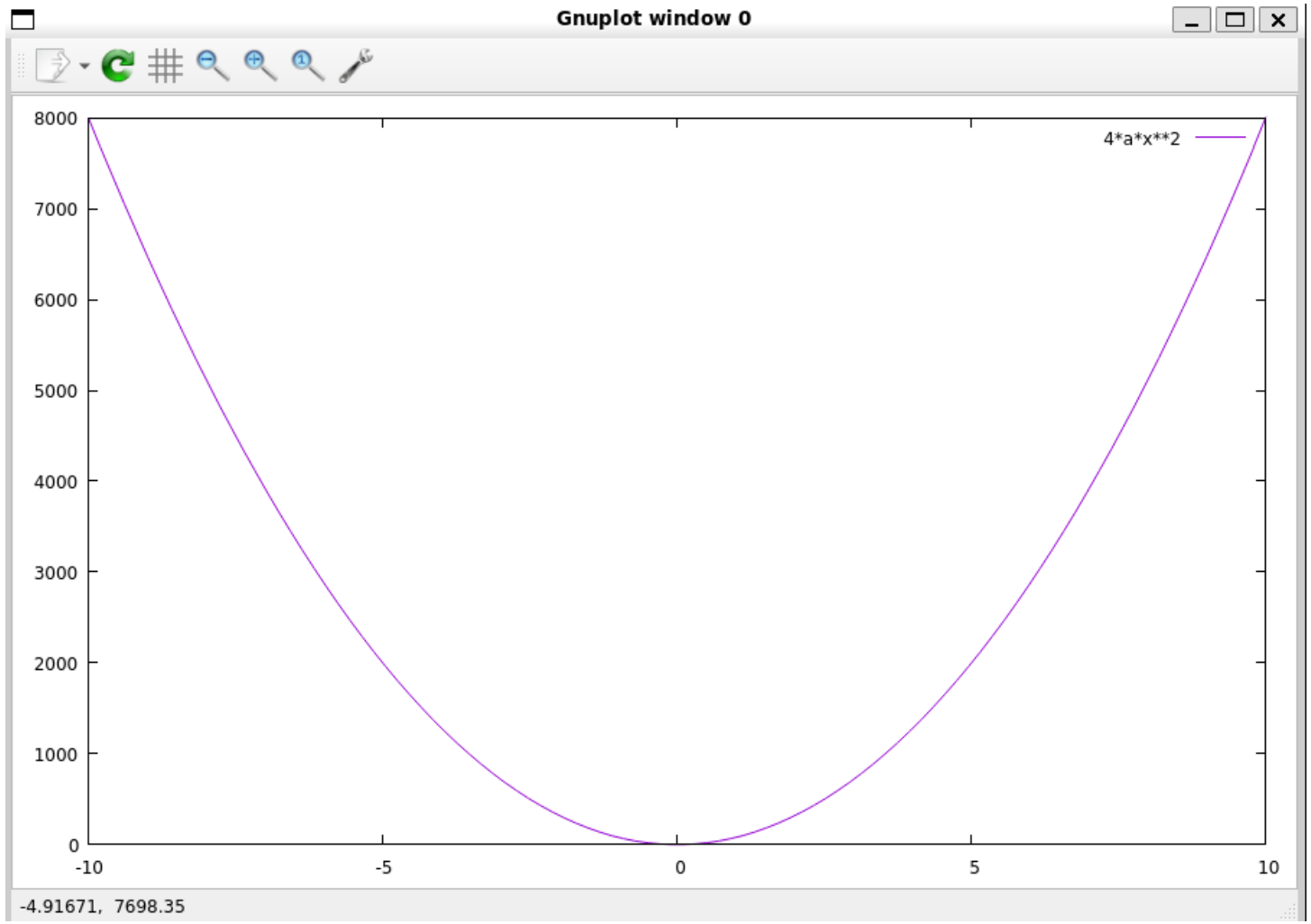
gnuplot>
```

- **Plotting a function:**
  - Plot function_with_variable
  - Example: plot sin(x)
  - Example: plot x**2



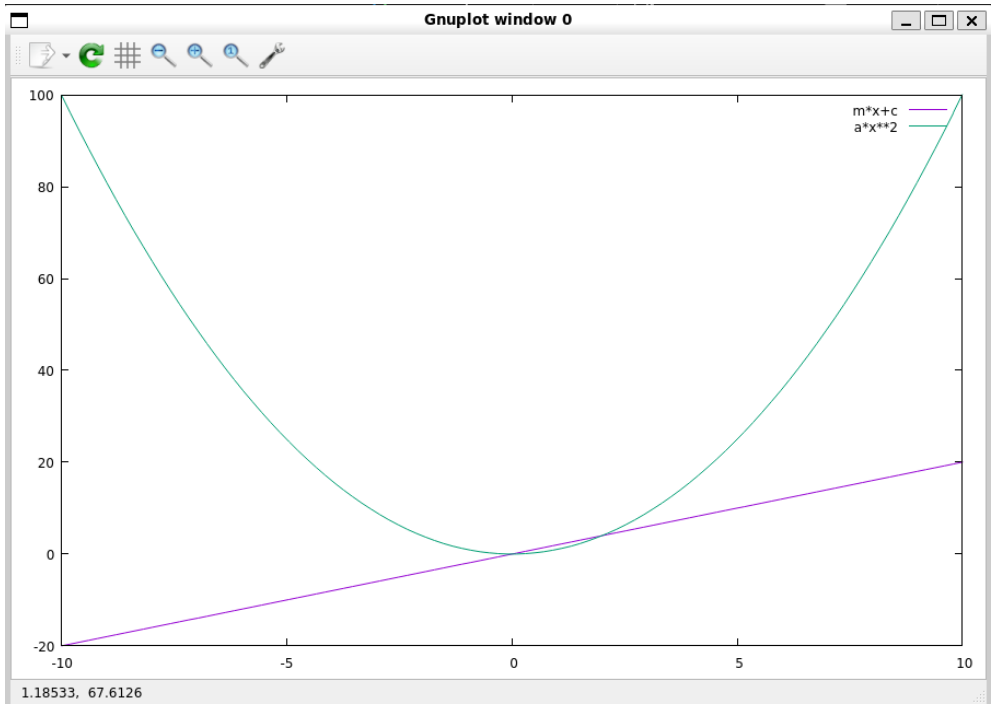
- **Plot y = $4ax^2$ where a = 8:**
  - a = 2
  - plot 4*a*x**2



## ❑ Replot:

- ➢ When we call the plot again after plot other function, the previous plot gets erased and the newly called plot is again generated.
- ➢ What if we want to plot second function while the first plot is present in the figure?
- ➢ The solution is replot.
- ➢ We can plot as many functions as we want in a single figure.

```
gnuplot> m=2
gnuplot> c=0
gnuplot> plot m*x+c
gnuplot> a=1
gnuplot> replot a*x**2
gnuplot>
```

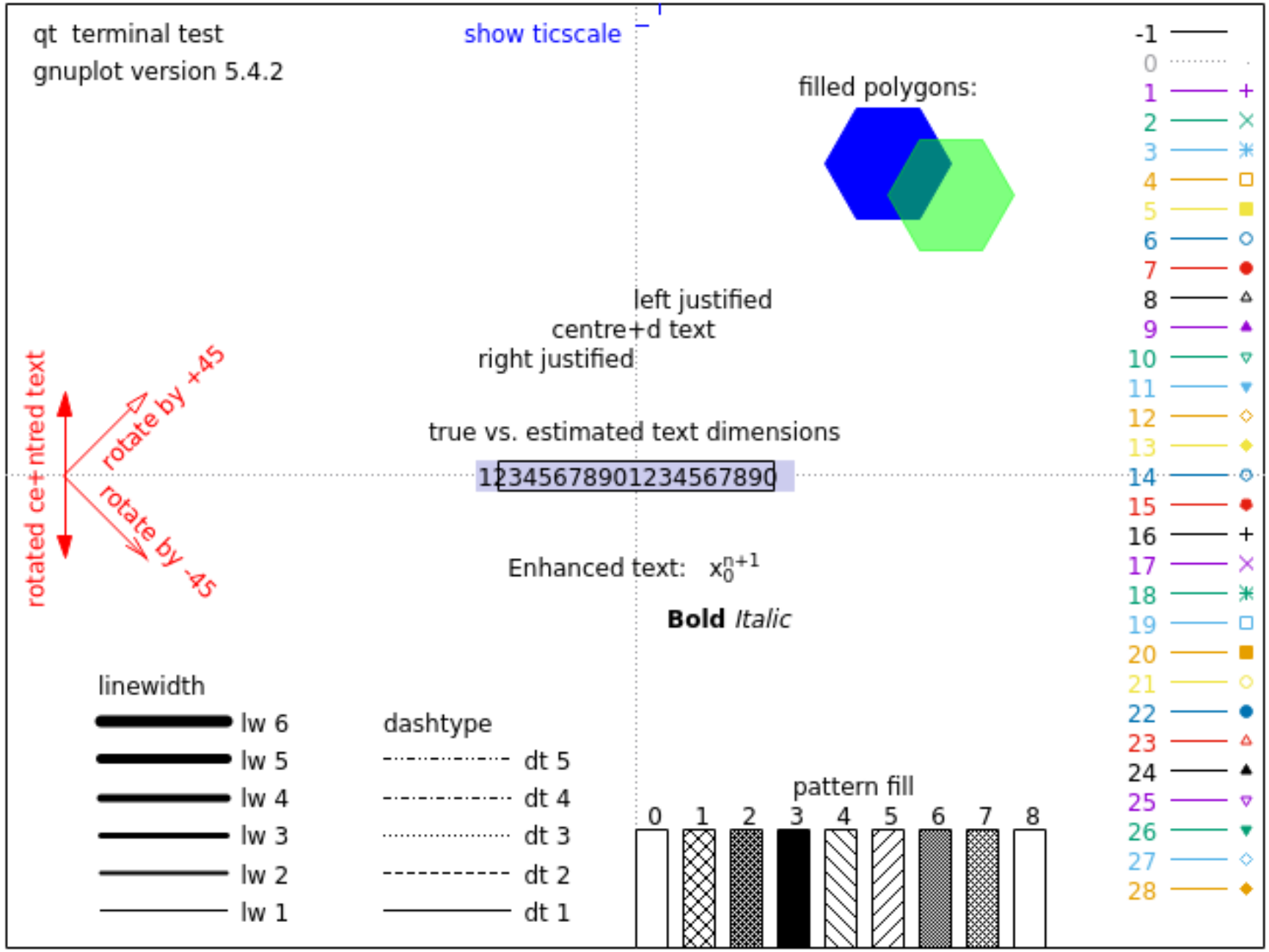


- **Setting Axis Ranges:**
  - **Set x-axis range:** set xrange [xmin:xmax]
  - **Set y-axis range:** set yrange [ymin:ymax]
- **Auto Scaling and Partial Ranges:**
  - Enable autoscaling: set autoscale
  - Partial Auto-scaling:
    - Fixed minimum x-value, auto-scale maximum: set xrange [0:*]
    - Auto-scale minimum, fixed maximum y-value: set yrange [*:<ymax>]
- **Resetting Ranges:**
  - unset xrange
  - unset yrange

- **Specifying Ranges in Plot:**
  - Set x-range only:  plot [-10:10] sin(x)
  - Set both x-range and y-range only:  plot [-10:10] [0:1] sin(x)
  - Set y-range only:  plot [] [0:1] sin(x)
- **Setting Margins:**
  - Left Margin: set lmargin <value>
  - Right Margin: set rmargin <value>
  - Top Margin: set tmargin <value>
  - Bottom Margin: set bmargin <value>
  - Display Margin Current Settings: show margin
  - Unsetting custom margin settings: use unset command

- **Title, xlabel, ylabel:**
  - set xlabel “Label Text” font “FontName, FontSize”
  - set ylabel “Label Text” font “FontName, FontSize”
  - set title “Title Text” font “FontName, FontSize”
- **Want Font in Bold or Italic:**
  - set xlabel “Label Text” font “FontName-Bold, FontSize” (Bold Font)
  - set xlabel “Label Text” font “FontName-Italic, FontSize” (Italic Font)
  - set xlabel “Label Text” font “FontName-BoldItalic, FontSize”
- **Setting Font Globally for all Texts:**
  - set term wxt font "Arial,14"

- **Line Styles available:**
  - lt 1: Solid, lt 2: Dashed, lt 3: Dotted ……
  - Example : plot sin(x) lt 2 title "Dashed Line"
- **Adjusting Line Colors:**
  - Example: plot cos(x) lc rgb "red" title "Red Line“ (Using color name)
  - Example: plot tan(x) lc rgb "#00FF00" title "Green Line" (Using hexadecimal code)
- **Adjusting Line Thickness:**
  - lw1: Default, lw 2: Medium, lw 3: Thick
- **Combining Style, Color, Thickness:**
  - plot cos(x) lt 2 lc rgb "blue" lw 2 title "Customized Line"


qt terminal test
gnuplot version 5.4.2
show ticscale
filled polygons:
left justified
centre+d text
right justified
rotated ce+ntred text
rotate by +45
rotate by -45
true vs. estimated text dimensions
12345678901234567890
Enhanced text: x0n+1
Bold Italic
linewidth
lw 6
lw 5
lw 4
lw 3
lw 2
lw 1
dashtype
dt 5
dt 4
dt 3
dt 2
dt 1
pattern fill
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